Dhanush Mathivanan

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Professional Summary

Data Science graduate student at The George Washington University, specializing in Data and Cloud Engineering, Machine Learning, Deep Learning, NLP, Generative AI, and large-scale data systems. AWS Certified cloud engineer with 9.8+ years of experience building distributed microservices, event-driven architectures, and scalable serverless data platforms on AWS for global enterprise systems, Fullstack web application development. Experienced in developing end-to-end machine learning pipelines and web applications using TensorFlow, Scikit-Learn, and FastAPI, Typescript, Nodejs, Angular, MySql, AWS, Terraform, Redis Cache, DynamoDb etc with strong expertise in predictive modeling, data analytics, software development and visualization. Passionate about applying machine learning and cloud-native technologies to solve large-scale real-world data problems.

Technical Skills

Programming: Python, SQL, TypeScript, JavaScript, R Studio, Linux, ActionScript

Machine Learning Frameworks: TensorFlow, Scikit-Learn, Hugging Face, Matplotlib, Seaborn, Polars, Pandas

Web Frameworks: Node.js, Express.js, Angular, FastAPI, Streamlit, Terraform, Electron, Redis, Ngrx Redux, Kafka

Databases: DynamoDB, MySQL, MongoDB, SQLite, DuckDb

Cloud & DevOps: AWS (Lex Bot, Connect customer, EC2, IAM, EventBridge, Lambda, CDK, S3, API Gateway, CloudFormation, Cognito, SQS/SNS, EMR), Jenkins, Docker, Github CI/CD

Web Technologies: HTML5, CSS3, JSON, XML, Bootstrap, jQuery

Testing & Debugging: AWS CloudWatch, Dynatrace, Mocha/Chai, Jasmine/Karma, Chrome DevTools, Dynatrace, Splunk

Version Control: Git, SVN

Tools: Tableau Desktop, Workbench, VS-Code, Atom, Pycharm, PySpark, Zoom Admin

Education

The George Washington University, Washington, DC
Master of Science in Data Science — Aug 2025 - May 2027
Coursework: Big Data Analytics, Cloud Computing, Machine Learning, Deep Learning, NLP, Data Warehousing

Professional Experience

The George Washington University, DC
Technical Support Assistant — Aug 2026 - Present

- Faced with manual, error-prone call routing across university IT support, engineered an AWS Connect/Lex/Lambda contact center with intent-based self-service, reducing tier-1 manual escalations by an estimated 42%.
- To eliminate manual website publishing and slow internal knowledge lookup, built an event-driven notification and AI knowledge pipeline using EventBridge, S3, Bedrock, and Amazon Q that auto-publishes updates on a recurring 15-minute schedule.
- Owned infrastructure reliability and device administration for 10,000+ users and 5,000+ network endpoints across Cisco CUCM, RedSky, centralized logs in Splunk Cloud to accelerate root cause analysis (RCA) on critical incidents.
- Led technical planning, testing, and cutover coordination for 2N IP intercom systems integrated with Cisco VoIP infrastructure, collaborating across facilities, security, and IT teams to ensure zero-disruption go-lives.

Probability Solutions LLC, VA
Data Science Intern — Jun 2026 - Aug 2026

- Conducted research on binary search tree algorithms and their applications in modern database technologies, evaluating their performance and scalability for data-intensive workloads.
- Developed software prototypes and performed statistical research analysis of tree-based algorithms using large-scale data inputs to evaluate computational performance and operational efficiency.

- Performed comparative analysis of different search tree implementations across CRUD operations, evaluating search, insertion, deletion, and update performance to identify algorithmic trade-offs.

Bosch Global Software Technologies
Client: MHP Porsche (Audi AG), Germany
Technical Project Specialist / Lead Engineer Mar 2024 - Aug 2025

- Built and deployed Porsche MBP workforce planning platform using AWS serverless microservices architecture with CDKTF, Lambda, API Gateway, DynamoDB, and Cognito, enabling scalable workforce data pipelines that support operational planning for a global automotive enterprise.
- Developed automated event-driven data ingestion pipelines by implementing JSON schema validation (JOI), S3-triggered Lambda processing, and API Gateway integrations, enabling real-time processing of enterprise HR datasets used for large-scale organizational planning decisions.
- Integrated legacy enterprise systems with modern cloud infrastructure through asynchronous event orchestration using SNS, SQS, and Lambda processing pipelines, improving reliability and enabling real-time synchronization of mission-critical workforce data across distributed systems.

Accenture Technologies
Client: Educational Testing Service (ETS), USA
Custom Software Engineering Team Lead Sep 2021 - Mar 2024

- Reduced candidate exam result processing time by 50% by architecting scalable AWS serverless data pipelines using Lambda, DynamoDB, Cognito authentication, SNS/SQS messaging, and CloudFormation infrastructure templates, improving throughput for a global educational assessment platform processing millions of exam records annually.
- Designed secure event ingestion architecture by validating candidate score events using AJV schema validation and processing events through SQS-triggered Lambda workflows, ensuring reliable real-time processing for high-volume exam scoring systems.
- Improved data platform efficiency and system reliability by developing automated DynamoDB maintenance scripts in Node.js and Python and implementing reusable TypeScript service components, enabling consistent API performance for enterprise-scale educational testing services.

HCL Technologies
Client: Roche Holding AG, Switzerland
Lead Engineer - Full Stack Cloud Development Nov 2020 - Sep 2021

- Accelerated healthcare diagnostic reporting workflows by 40% by developing RESTful APIs and reporting services using Node.js, Express.js, Angular, and MySQL, enabling faster processing of patient diagnostic data for Roche's global healthcare operations.
- Automated enterprise reporting pipelines by implementing dynamic PDF and Excel report generation using Node.js libraries (ExcelJS, PDF-node) and streaming data pipelines, reducing manual reporting overhead across laboratory testing workflows.
- Enabled secure healthcare system integrations by implementing OAuth/JWT authenticated REST APIs and AWS S3-based storage pipelines, ensuring reliable and compliant exchange of medical workflow data across distributed clinical systems.

Object Frontier Software Pvt Ltd
Client: International Flavours Fragrance (IFF), USA
Software Engineer - Full Stack Development Aug 2018 - Nov 2020

- Developed offline-first enterprise desktop application using Electron framework with embedded Node.js server and SQLite/MySQL synchronization architecture, enabling uninterrupted operations for global manufacturing and supply chain workflows.
- Improved application scalability and user transaction management by implementing Angular ngrx-redux state architecture with reducer-action data flow patterns, enabling advanced features such as undo/redo operations across complex enterprise transactions.
- Strengthened backend reliability and database performance by integrating Node.js microservices with MySQL through Sequelize ORM and implementing structured logging using Winston, improving system observability and production stability for enterprise-grade systems.

V2Soft Pvt Ltd
Client: Fiat Chrysler Automobiles (FCA), USA
Software Engineer - HMI S/W Development Oct 2016 - Aug 2018

- Developed responsive automotive Human-Machine Interface dashboards using Angular, HTML5, Bootstrap, and modular JavaScript components, improving usability across vehicle infotainment and diagnostic systems.
- Enabled scalable vehicle telemetry data processing by implementing MongoDB batch jobs and Node.js backend APIs, supporting ingestion and analysis of automotive diagnostic logs across vehicle platforms.
- Implemented real-time event communication by developing Socket.IO-based messaging between backend Node.js services and frontend dashboard modules, enabling real-time monitoring of vehicle operational data.

Certifications & Awards

Phi Delta Gamma Fellowship recipient - George Washington University (2026)

Global Leaders Fellowship recipient - George Washington University (2025)

Introduction to R (DataCamp) (2025)

AWS Certified Solutions Architect, Associate - SAA-C03 (2024)

AWS Certified Developer, Associate - DVA-C02 (2023)

Client Value Creation Award, Best People Award - Accenture (2022)

Data Analysis Projects

- The Enterprise Customer Risk Intelligence System based on US Consumer Finance Data (2026): Code Base Github
- Heart Disease prediction based on US CDC Survey Data (2024): Code Base Github , ML Trained Predictor App
- Health Status prediction based on Income factors in US Population CDC Survey Data (2024): Code Base Github